BY ORDER OF THE COMMANDER
ELLSWORTH AFB



AIR FORCE INSTRUCTION 13-212

AIR COMBAT COMMAND
Supplement

ELLSWORTH AIR FORCE BASE

Addendum A

8 MAY 2012

Space, Missile, Command, and Control

RANGE PLANNING AND OPERATIONS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

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(Lt Col Jason R. Combs)
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This addendum complements **AFI 13-212 dated 16 Nov 2007**, and **AFI 13-212_ACC Sup 1 dated 29 April 2010**. This addendum establishes procedures and policies applicable to all agencies using Belle Fourche Electronic Scoring Site. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual 33-363, *Management of Records*, and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS) located at **https://www.my.af.mil/gcss-af61a/afrims/afrims/**. Send comments and suggested changes to this supplement through channels, using AF Form 847, *Recommendation for Change of Publication*, to 28 OSS/OSOA, 1956 Scott Dr, Suite 201, Ellsworth AFB SD 57706-4710.

Chapter 1

RESPONSIBILITIES

1.1. General Information: The 28th Bomb Wing (28 BW) is designated as the Range Operating Authority (ROA) for Belle Fourche Electronic Scoring Site. IAW AFI 13-212, ACC Sup 1, the 28th Bomb Wing Commander has delegated the ROA responsibilities to the 28th Operations Group Commander.

1.2. 28 BW ROA: The ROA will ensure all squadrons and units operating with or providing support to Belle Fourche Electronic Scoring Site (ESS) comply with this addendum.

1.3. Other Agencies:

1.3.1. 28th Logistics Readiness Squadron (28 LRS)

1.3.1.1. Will perform safety and road worthiness tests for Radar systems prior to shipping between government agencies IAW local letters of agreement.

1.3.1.2. The 28 LRS Traffic Management Office will provide shipping and receiving services for Belle Fourche ESS.

1.3.1.3. 28 LRS Supply side will provide support for all National Stock Number items.

1.3.2. 28th Communications Squadron

1.3.2.1. Will provide communications support for voice and data.

1.3.2.2. Will provide frequency management assistance.

1.3.2.3. Will facilitate obtaining communications engineering support.

1.3.2.4. Will provide network troubleshooting,

1.3.2.5. Will provide Automated Data Processing Equipment services.

1.3.3. 28th Medical Group

1.3.3.1. Will provide water and soil sampling.

1.3.3.2. Will provide Radiation Hazard safety assessments for new or altered radar systems.

1.3.4. 28th Maintenance Squadron

1.3.4.1. Will provide Test Measurement Diagnostic Equipment/Calibration services.

1.4. Host Unit: The 28th Operations Support Squadron (28 OSS) provides operational control and oversight of the ESS through two full time civilian employee positions which are designated as 28th Operations Support Squadron Operating Location Alpha (28 OSS/OL-A).

1.5. 28th Civil Engineering Squadron

1.5.1. Will provide maintenance and repairs to range facilities which are beyond the capability of the range contractor or out of scope of the Air Combat Command (ACC) range contract as determined by Headquarters Air Combat Command Acquisition Management and Integration Center.

1.5.2. Will provide environmental inspections and assessments for Government owned/leased range property.

1.5.3. Will provide annual Fire, Safety, and Real Property inspections annually.

1.5.4. Will provide utility services (negotiations, funding, and bill processing) for Air Force owned and/or leased Real Property.

1.5.5. Will provide Real Property services for Air Force owned or leased property.

1.6. Weather

1.6.1. The 25th Observation and Weather Service (Davis-Monthan AFB, AZ) provides weather alerts to the range based on radar equipment operating parameters and personnel safety IAW Military Operating Area (MOA) 15-75, dated 9 Jan. 05.

1.7. Range User

1.7.1. Primary range users include 28 BW (Ellsworth AFB, SD), 5th Bomb Wing (Minot AFB, ND) and 55th Recon Wing (Offutt AFB, NE).

1.7.2. Occasional users include 7th Bomb Wing (Dyess AFB, TX) and 2nd Bomb Wing (Barksdale AFB, LA).

1.8. Unit Feedback

1.8.1. Range feedback should be provided to the Range Manager/Range Operations Officer at DSN 675-6050.

1.9. Scheduling Authority

1.9.1. Scheduling Authority for the Belle Fourche ESS is 28th Operations Support Squadron Current Operations Flight (28 OSS/OSO), Ellsworth AFB, SD. DSN 675-5371.

Chapter 2

DESCRIPTION OF RANGE AND MILITARY OPERATING AREA

2.1. General Information: Belle Fourche ESS is an ACC Primary Training Range located within the Powder River MOA in Western South Dakota, North Eastern Wyoming, and South Eastern Montana. The ESS is operated under ACC Contract # FA 4890-07-C-0016 by civilian contractors.

2.1.1. Capabilities: The Belle Fourche ESS specializes in simulated Electronic Combat (EC) training, Infrared (I/R) targeting, and Laser targeting and detection training. EC capabilities include realistic simulated Surface to Air and Air to Air threat environments that include analysis, feedback, and debriefing capabilities. Belle Fourche ESS provides high fidelity threat signals to aircrews while maintaining the flexibility to meet specific crew requirements. Pre-planned scenarios, individual threat system simulation, individual signals or site options may be requested. The ESS operates 6 individual heated target sets (3xSA-6 TELs, 1 SCUD-B, 1 ZSU-23/4, and 1 BMP-1) and two target arrays consisting of civilian vehicles (cars, trucks, and motorcycles) which are heated for I/R signatures and are authorized for EYE-SAFE lasing. Coordinates for all target sets may be obtained at:
https://afkm.wpafb.af.mil/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81
https://www.my.af.mil/afknprod/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81

2.1.2. Hour of Operation: The site is vulnerable for 60 hours per week as identified in the Operations and Maintenance contract and as defined by the 28 OSS. The vulnerability hours are reviewed and subject to change on a quarterly basis. Requests for change to the site's vulnerability hours should be directed to the Belle Fourche ESS Range Management office at DSN 675-6050. The current hours of operation are posted at:

https://afkm.wpafb.af.mil/ASPs/CoP/OpenCoP.asp?Filter=OO-Op-AC-81
https://www.my.af.mil/afknprod/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81
Requests for additional operational hours or hours outside of the published vulnerability hours may be made to the Range Management Office at DSN: 675-6050. Approvals of these hours are normally reserved for special missions or normal missions with extenuating circumstances. 72 hour notice to the contractor is required to assure approval.

2.1.3. Scheduling Procedures: 28 OSS/OSO is the scheduling authority for the Powder River MOA and the Belle Fourche ESS. All scheduling of activity for either of these resources should be requested through this authority at DSN: 675-5371. Specific instructions for scheduling and pre-run information for the Belle Fourche ESS are addressed in Chapter 4, Para. 4.2.

2.2. Restrictions:

2.2.1. Common/General: Chaff, Flares, and Smokey Surface to Air Missiles are restricted and may not be used in the Powder River MOA.

2.2.2. Range: All Lasing on targets in the approved target sets (MM-8 & MM-11) must be in the EYE-SAFE /TRAINING Mode only.

2.2.3. Airspace: Available training airspace includes two MOAs: Powder River A and B, four Air Traffic Control Assigned Airspaces (ATCAAs): Powder River, Gateway, Black Hills West, and Crossbow, and three Military Training Routes (MTRs): IR-473, IR-485, and IR-492. All are maintained by the 28th Operations Support Squadron Airspace Management (28 OSS/OSOA) and information is posted on the web:

https://afkm.wpafb.af.mil/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81or
https://www.my.af.mil/afknprod/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81
Powder River A/B MOAs are defined in FLIP AP/1A, the associated ATCAAs are defined in a Letter of Agreement (LOA) with Denver Air Route Traffic Control Center (ARTCC), and the MTRs are defined in FLIP AP/1B. Scheduled activity for the ESS does not include any of the training airspace identified here. Associated Airspace is scheduled separately through 28 OSS Scheduling (DSN 675-5371, COM 605-385-5371).

2.3. Range and MOA. Belle Fourche ESS is an electronic range only. Weapons releases are not authorized. Belle Fourche ESS is located beneath Powder River MOAs, their associated ATCAAs, and three MTRs (see 2.2.3 for references to detailed airspace data).

2.3.1. Air-To-Ground Range: N/A

2.3.2. Tactical Range: N/A

2.3.3. Electronic Range: The Electronic Scoring Site presents a dense electronic threat environment consisting of 10 remote independent threat emitters capable of presenting at least one threat family each. The main site operates a multiple threat emitter system capable of representing multiple threats including ground based (surface to air), (air to air), and Naval threats. Mission feedback and debriefing products can be provided through the Threat Reaction and Analysis Indicator System (TRAINS) and Range Instrumentation and Integration System (RIIS). A Battlefield Communications Simulator system is scheduled to be installed and available to produce multiple radio transmissions (Fog of War) by the publication date of this addendum.

2.3.4. Air-To-Air Range: N/A

2.3.5. MOA: See 2.2.3

2.4. Range Routes, Air Refueling Tracks and Remotely Piloted Aircraft (RPA) Corridors: N/A

2.5. Landing Zones and Drop Zones: N/A

Chapter 3

OPERATIONS/WEAPONS DELIVERY PROCEDURES

3.1. Overview: Belle Fourche ESS is a no drop range and weapons delivery is not authorized. Limited Eye-Safe Lasing is authorized at two locations within the MOA (MM-8 & MM-11). I/R enhanced targets are available and co-located with the remote threat emitters.

3.2. Authorized Ordnance: None

3.3. Restrictions, Limitations and Footprint Data: N/A

3.4. Laser or Directed Energy Operations: Laser designation is authorized in the Eye-Safe mode at MM-8 and MM-11 at altitudes above 5,000' Above Ground Level. No over the horizon or Aircraft to Aircraft lasing is authorized.

3.4.1. Laser Systems: N/A

3.4.2. Airborne: Any US Military Targeting POD in Eye-Safe Mode.

3.4.3. Ground Laser Operations: N/A

3.4.4. Directed Energy Operations: N/A

3.5. Night Operations: Night operations at Belle Fourche ESS are limited to Electronic Attack/Electronic Combat operations.

3.5.1. Night Lighting: None

3.5.2. Night Vision Detection: N/A

3.5.3. Infrared Enhanced Targets: The ESS operates 6 full sized heated threat mock-up targets including 3 SA-6 TELs, 1 ZSU-23/4, 1 SCUD-B, and 1 BMP-C. The ESS also operates 2 target sets consisting of 6-8 civilian type vehicles (cars, trucks, and motorcycles) in each target set with up to 4 vehicles heated in each set.

3.6. Unmanned Threat Emitters: The ESS operates 10 remotely located independent unmanned threat emitters within the Powder River MOA and represents a dense threat environment. All remote threat emitters use Identification, Friend or Foe (IFF) for tracking. Each remote threat represents/simulates at least one threat, i.e., SA-2, SA-6, SA-8, etc., depending on the model. The ESS currently operates B, C, G, H, and K models.

3.7. Transition corridor operations: N/A

3.8. Helicopter operations: N/A

3.9. Weather: Belle Fourche ESS receives point weather warnings IAW Chapter 1, Para. 1.6. Some or all of the range threat emitters and/or communications equipment may be shut down due to forecasted weather events that represent a hazard to the operators or the equipment. The contractor has the authority to close the site due to severe weather and is directed to shut down threat systems when weather (wind, snow, ice) exceed Technical Order limits. Reporting site closure to the 28 OSS and the 28 BW Command Post is required. The range does not have the authority to close the MOA.

3.10. Minimums and Fouls: N/A

3.11. Emergency Procedures:

3.11.1. Emergency Airfields: See attachment 2 for Auxiliary/Emergency Airfields.

3.11.2. Dropped Object and Inadvertent Release: N/A

3.11.3. Hung Ordnance and Unsafe Gun Procedures: N/A

3.11.4. Jettison Procedures: N/A

Chapter 4

ELECTRONIC COMBAT RANGE/ELECTRONIC SCORING SITE

4.1. Purpose: Belle Fourche ESS provides Electronic Warfare (EW) and Target Acquisition training to military aircrews. The ESS simulates hostile ground based air defense threats and provides target mock-ups for visual identification. The primary purpose of the ESS is to provide electronic threat recognition and targeting training to United States and allied aircrews.

4.2. Scheduling: All scheduling for the Belle Fourche ESS is done through the 28th Operations Support Squadron Flight Records (28 OSS/OSOS) office at DSN: 675-5371. The Range operators are not authorized to de-conflict scheduling issues. A Pre-Run information sheet is available at **http://nellis.ahntech.com/** and should be used and Faxed to DSN 675-2243 to ensure proper run coordination. Specific aircrew EC and heated target requests may also be accomplished directly by calling the ESS at DSN 675-2241/2242. The ESS monitors the schedule for changes throughout the range operating hours.

4.3. Events: Aircrews may request pre-planned scenarios, individual threat systems, individual signals, or site options on each engagement along with options for Battlefield Communications Simulation traffic, and/or heated target activation.

4.4. Restrictions: See Para. 2.2

4.5. Communications: Belle Fourche ESS operates on a primary Ultra-High frequency of 381.1 Megahertz with a call sign of Belle Fourche Bomb Plot.

4.6. Tactics: No Tactics are authorized.

4.7. Operations: The ESS is available to provide activity 60 hour per week as published (see Chapter 2, Para. 2.1.2). Special circumstances/missions which require ESS activity outside of the published hours must be coordinated with range management at DSN 675-6050. 72 hours advance notice is required to assure contractor availability/approval. EW requests received from unscheduled activity (Bootleg) will be honored providing it does not interfere with scheduled activity.

4.8. Simulators: Belle Fourche ESS Primary Threat simulators are listed below.

4.8.1. AN/MST-T1A Multiple Threat Emitter (MUTES): This manned threat simulator is capable of presenting up to 5 threat signals simultaneously from a signal inventory of over 100 signals. The threat is an IFF tracker. Belle Fourche ESS operates one MUTES which is co-located at the main scoring site.

4.8.2. AN/MST-T1V Mini-MUTES: This unmanned remotely located threat emitter system made up of a Master Control Group and up to 5 remote emitters. Each remote emitter represents an individual threat system. The system is an IFF tracker. Belle Fourche ESS operates two Mini-MUTES systems with 10 remote threats located within the Powder River MOA. Mini-MUTE threat locations can be found at

https://afkm.wpafb.af.mil/ASPs/CoP/OpenCoP.asp?Filter=OO-Op-AC-81 or
https://www.my.af.mil/afknprod/ASPs/CoP/OpenCoP.asp?Filter=OO-OP-AC-81

4.8.3. Battlefield Communications Simulator System (BCSS): The BCSS is scheduled to be in place at Belle Fourche ESS in Nov 2010 and will provide Fog of War transmissions to simulate battlefield conditions. The simulations may be pre-recorded or live broadcasts over various frequencies.

4.9. Feedback: The RIIS debriefing program is available on the individual squadrons Integrated Tactics Assessment System computer. Belle Fourche ESS can provide Post Mission reports with threat signal on/off times and Electronic Countermeasures analysis with pre-coordinated requests.

Chapter 5

RANGE CONTROL OFFICER (RCO) PROCEDURES

5.1. Responsibilities: Belle Fourche ESS is a No-Drop range and no RCO is assigned/ required. This chapter is retained for future use.

Chapter 6

AIR COMBAT MANEUVERING INSTRUMENTATION (ACMI)

6.1. Purpose: Belle Fourche ESS does not have ACMI capability. This chapter is retained for future use.

JEFFREY B. TALIAFERRO, Colonel, USAF
Commander, 28th Bomb Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References:

AFI 13-212_ACCSUP 1; Range Planning and Operations, 29 Apr 2010

ACCI 10-707, ACC Electronic Attack Training and Emissions Control Procedures, 4 Aug 2011

25 OWS Memorandum of Agreement, 15-75, 9 Jan 05

AF Form 847, Recommendation for Change of Publication

Abbreviations and Acronyms:

28 BW——28th Bomb Wing

28 LRS——28th Logistics Readiness Squadron

28 OSS——28th Operations Support Squadron

28 OSS/OL-A——28th Operations Support Squadron Operating Location Alpha

28 OSS/OSOA——28th Operations Support Squadron Airspace Management

28 OSS/OSOS——28th Operations Support Squadron Flight Records

AI——Airborne Intercept

ASIN——Airspace Scoring Information Network

Bootleg Activity——Unscheduled EC/RBS Activity

CBDA——Combat Battle Damage Assessment

IFF——Identification, Friend or Foe

ITAS——Integrated Tactics Assessment System

I/R——Infrared

MAMS——Military Airspace Management System

MUTES——Multiple Threat Emitter

ONHO——Outside Normal Hours of Operations

RBS——Radar Bomb Scoring

RF——Radio Frequency

RIIS——Range Integration Instrumentation System

SEMR——Senior Executive Management Review

TRAINS——Threat Reaction and Analysis Indicator System

Vulnerability——Indicates a period of time when the ESS is manned and required to conduct EC/RBS activity

Terms

Emitter/Simulator—Generic terms used to describe threat equipment operated at Electronic Combat Ranges (ECR) and Electronic Scoring Sites (ESS). However, ACC operates a variety of equipment including, but not limited to: Emitter only systems, Emitter-Receiver-Processors, and replica type systems. Carefully compare the similarities and features of different systems and consider operational requirements. Surface-to-Air Missile (SAM) and Anti-Aircraft Artillery (AAA) simulators include all manned and unmanned threat emitters.

Government Controlled—Control exercised by any agency of the federal government, not just USAF or DoD.

Major Command (MAJCOM)—A major subdivision of USAF assigned a major part of the Air Force mission. Major commands report directly to Headquarters United States Air Force (HQ USAF). The ANGRC/DO serves as the MAJCOM for Air National Guard ranges.

Military Operations Area (MOA)—Special Use Airspace allocated to the military to separate/segregate certain military activities from Instrument Flight Rules (IFR) traffic, and to identify the location of these military activities to Visual Flight Rules (VFR) traffic. VFR aircraft are not restricted from transiting MOAs.

Military Training Route (MTR)—A low-level, high-speed training route established IAW criteria in FAA Handbook 7610.4, *Special Military Operations*. MTRs are used by DoD to conduct low altitude navigation and tactical training, in instrument and visual weather conditions, below an altitude of 10,000 feet MSL and at airspeeds more than 250 KIAS. Routes are established as IFR routes (IR) or VFR routes (VR). The FAA has approval authority to implement IRs and the appropriate MAJCOM approves VR implementation. Environmental documentation is required for implementation IAW AFI 32-7061. VRs are processed through the FAA via the AFREP. MTRs are published in FLIP AP/1B and charted on FAA Sectionals and DoD Low IFR charts. AFREPs assign all route numbers.

Proponent—Any office, unit, or activity that proposes to initiate an action.

Range—Military—Designated land, and water areas set aside, managed, and used to research, develop, test, and evaluate military munitions, other ordnance, or weapons systems, or to train military personnel in their use and handling. Ranges include firing lines and positions, maneuver areas, firing lanes, test pads, detonation pads, Target Areas, and Hazard areas. It includes the airspace above the range.

Range Operating Agency (ROA)—The agency designated to operate and maintain the range. The ROA may delegate the daily scheduling, management, and maintenance of the range to any appropriate subordinate unit.

Range Integration Instrumentation System (RIIS)—The primary mission of the RIIS is to support the ACC policy of sustained combat readiness by providing feedback in the form of graphic debriefing products. RIIS integrates all ESSs as well as some ECRs.

Scheduling Authority—The agency, organization or military activity responsible for scheduling all activities in designated land and airspace areas.

Scoping—A public process for proposing alternatives to be addressed and for identifying significant issues related to a proposed action.

Shared Use—When participating (as defined by the using agency) and non- participating (civil or military) users share designated land and/or airspace areas on a noninterference basis.

Sortie—A term to describe a single training event performed by one aircraft from takeoff through landing. A single aircraft may only log one sortie per flight.

Sortie Operation—A term to define the use of training airspace or ranges by a single aircraft. A single aircraft may log multiple sortie operations per flight by using different training areas or the same area several times throughout the same flight.

Special Use Airspace (SUA)—Airspace of defined vertical and lateral dimensions wherein activities are confined. Certain limitations or restrictions may be imposed on non-participating aircraft. Except for Controlled Firing Areas, SUA is depicted on aeronautical charts. Types of Special Use Airspace include:

Alert Area.—Controlled Firing Area.

Military Operations Area.—Prohibited Area.

Restricted Area.—Warning Area.

Attachment 2

AUXILIARY/EMERGENCY AIRFIELDS

Ellsworth AFB, SD (KRCA)

WGS-84: N44-09 W103-06

Field Elevation: 3280 MSL

Runway: 13,497 x 300 feet

Location from main site: 65 NMs on 123 degree radial.

Rapid City Regional Airport, SD (KRAP)

WGS-84: N44-03 W103-03

Field Elevation: 3204 MSL

Runway: 8701 x 150 feet

Location from main site: 70 NMs on 126 degree radial.

Spearfish/Black Hills-Clyde Ice Field (KSPF)

WGS-84: N44-29 W103-47

Field Elevation: 3931 MSL

Runway: 5498 x 75 feet

Location from main site: 30 NMs on 135 degree radial.

Gillette-Campbell County (KGCC)

WGS-84: N44-21 W105-32

Field Elevation: 4363 MSL

Runway: 7500 x 150 feet

Location from main site: 65 NMs on 231 degree radial.

Miles City-Frank Wiley Field (KMLS)

WGS-84: N46-26 W105-53

Field Elevation: 2628 MSL

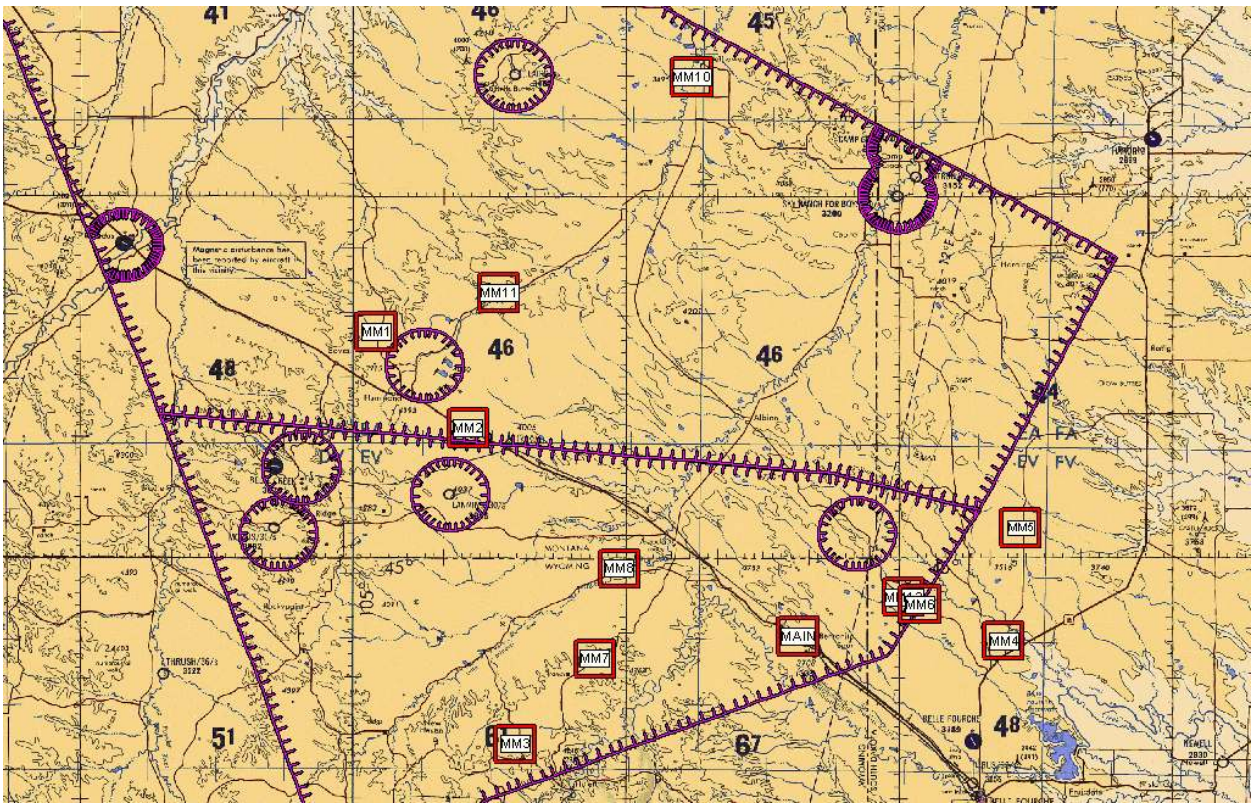
Runway: 5680 x 75 feet

Location from main site: 115 NMs on 312 degree radial.

Attachment 3

RANGE LAYOUT

Sites



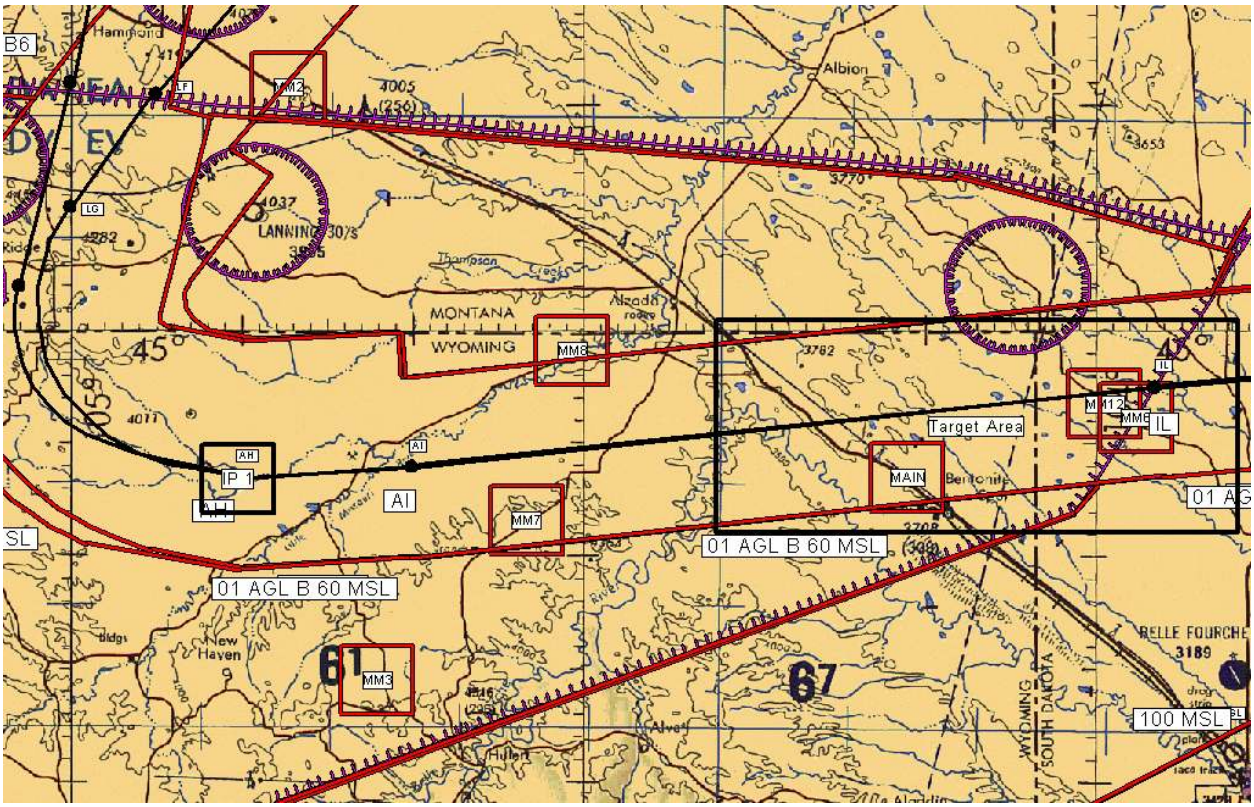
Attachment 4

FLIGHT PROFILES

IP/Target Combinations

Low Altitude:

IP 1 (N44-53.5 W 104-50.0) to Targets 3, 4, 5, 11, 13, 17, 18



High Altitude:

IP 8 (N44-39.0 W102-00.0) to Targets 14, 15, 16, 18

IP 9 (N44-55.0 W105-50.0) to Targets 14, 15, 16, 18

